

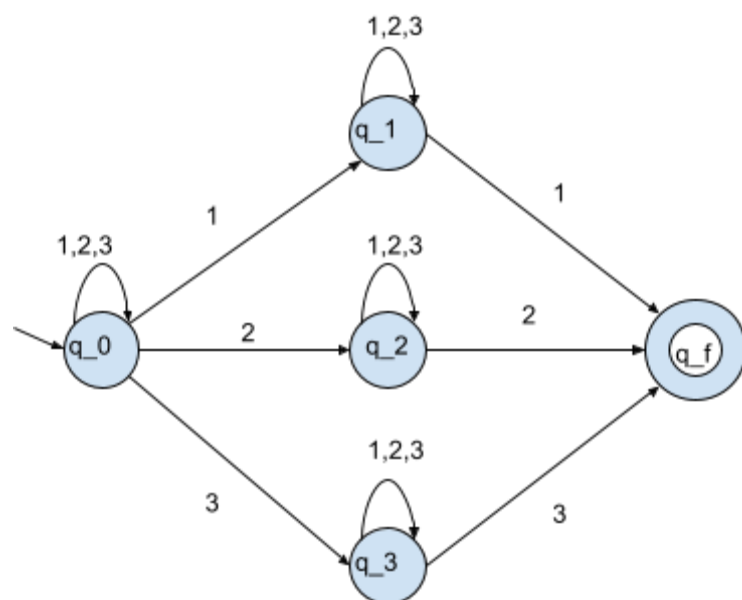
FINITE AUTOMATA (FA)

1. Given the FA: $M = (Q, \Sigma, \delta, q_0, F)$, $Q = \{q_0, q_1, q_2, q_3, q_f\}$, $\Sigma = \{1, 2, 3\}$, $F = \{q_f\}$,

δ	1	2	3
q_0	$\{q_0, q_1\}$	$\{q_0, q_2\}$	$\{q_0, q_3\}$
q_1	$\{q_1, q_f\}$	$\{q_1\}$	$\{q_1\}$
q_2	$\{q_2\}$	$\{q_2, q_f\}$	$\{q_2\}$
q_3	$\{q_3\}$	$\{q_3\}$	$\{q_3, q_f\}$
q_f	$\emptyset$	$\emptyset$	$\emptyset$

Prove that $w = 12321 \in L(M)$

Sol.: B: XXXXXXXXXXXXXXXXXXXX



$$(p, a^{k+1}) \mid - (p, a^k) \mid - (p, \varepsilon) \Rightarrow (p, a^{k+1}) \mid - (p, \varepsilon) \Rightarrow P(k+1) \text{ -true}$$

Ind. hyp.

Similarly, we demonstrate b.

2. ? $L(M) \subseteq L$ (M does not accept anything else but sequences of that shape)

In order to reach the final state q from the initial state p , we should read at least one b . Before the mandatory b , we can read any natural number of a 's, while remaining in state p , and after the mandatory b we can read any natural number of b 's, while remaining in state q . Therefore, M accepts only sequences of the shape $a^n b b^k$, $n, k \in N$

Obs. In order for such a reasoning to count as proof, you should make sure that you have covered all paths from initial state to final states.

3. Build FAs that accept the following languages

- a. Integer numbers
- b. Variable declarations (Pascal, C, ...)
- c. $L = \{0^n 1^m 0^q \mid n, m \in N^*, q \in N\}$
- d. $L = \{0(01)^n \mid n \in N\}$
- e. $L = \{c^{3n} \mid n \in N^*\}$
- f. The language over $\Sigma = \{0, 1\}$ having the property that all sequences have at least two consecutive 0's.

a. IW -> 